

To Balance the Mechanical Toughness and Electrical Insulation of Polypropylene by Blending and Grafting Modifications

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Abstract: To balance the mechanical toughness and electrical insulation of polypropylene (PP) for recyclable cable insulation application has attracted increasing attention. Due to the different molecular conformation, isotactic polypropylene (IPP) always deliver excellent electrical insulation but poor mechanical toughness, while block polypropylene (BPP) has been dramatically opposed properties. Blending IPP with BPP at an appropriate content is an efficient modification to reconcile the toughness and insulation. In this study, we firstly investigate the blending ratio of IPP

and BPP, and found that 60 wt% : 40 wt% is the optimized for achieving outstanding overall performance. Furthermore, polyolefin elastomer (POE) and styrene-grafted POE (POE-g-St) have been incorporated into IPP/BPP blends respectively to intensify the toughness as well as the insulation property. The microstructure, electrical properties, mechanical properties, and thermal properties of the composites are systematically analyzed and discussed. The results demonstrate that the composite with a grafting ratio of 1.21 wt% styrene exhibits superior overall performances, the breakdown field strength of 314.02 kV/mm and elastic modulus of 352.54 MPa have been achieved. Additionally, the modified composites also possess a high melting temperature, high volume resistivity and excellent space charge suppression capability, achieving a synergistic effect of electrical, mechanical and thermal properties.

Keywords: polypropylene, blends, toughness, insulation, graft modification.

1 Introduction

High-voltage DC transmission has the advantages of large transmission capacity, long power delivery distance and low loss, in which power cables, play a key role in the transmission and distribution system. Cross-linked polyethylene (XLPE) is widely used because of its good dielectric and mechanical properties and has become the most commonly used insulation material for power cables. However, with the widespread use of XLPE cable materials, their disadvantages have gradually become

apparent. On the one hand, XLPE cables need to be cross-linked in the manufacturing process, which consumes a lot of energy; on the other hand, XLPE cables are thermosetting materials due to their molecular network structure, which can't be recycled and only be incinerated or buried after the cables are decommissioned, resulting in the environment pollution^[1-3]. With the strengthening of environmental protection concepts and the proposal of the "Carbon Peaking and Carbon Neutrality Goals" strategy in recent years, it is imperative to develop environmentally friendly and recyclable power cable materials.

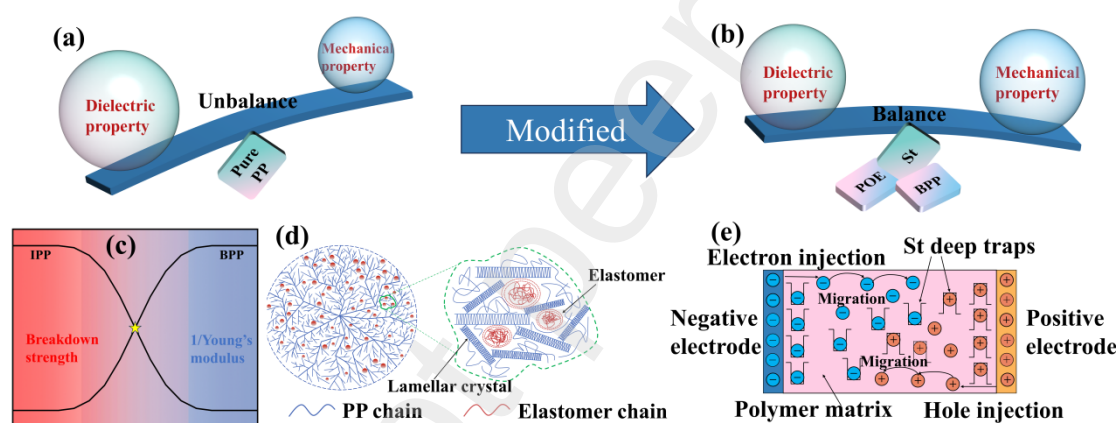


Fig. 1 (a)Unbalance of dielectric and mechanical properties (b)Balance of dielectric and mechanical properties (c) Characteristics of IPP and BPP. (d) Microscopic schematic of the modified PP. (e) Mechanism of charge transport

In recent years, polypropylene (PP) cable materials have attracted considerable attention. Compared with XLPE, polypropylene does not require cross-linking during the production process, which simplifies the process and reduces energy consumption and costs. Polypropylene is a thermoplastic material that can be recycled and reused after retirement, avoiding environmental pollution and waste of resources. Polypropylene has a high melting point, can withstand high operating temperatures, has a high breakdown field strength and volume resistivity, which is significant for

improving the cable voltage operating level and carrying capacity^[4,5]. However, conventional polypropylene has high crystallinity, high rigidity, low flexibility and poor mechanical properties, which should be improved for cable laying process and engineering application.

Many previous studies have reported that the modifications for reconciling the electrical insulation and mechanical toughness of polypropylene, including blending, incorporating elastomer and molecular chains grafting. Hosier et al. blended isotactic polypropylene (IPP) and block polypropylene (BPP) for maintaining a balance between the electrical and mechanical properties of polypropylene. The results showed that the blending PP containing 50% IPP possessed much higher breakdown field strength and thermomechanical properties comparable to XLPE^[6]. Gao et al. blended IPP and propylene-based elastomer (PBE) at different weight percentages. The results showed that the PP/PBE composite introduced deep traps, which gave it a higher breakdown field strength, especially PP and PBE had good compatibility and exhibited excellent mechanical properties in terms of elongation at break and tensile strength^[7]. Zhou et al. investigated the microstructure and mechanical properties of polypropylene/polyolefin elastomer (PP/POE) blends. As the elastomer POE content increased, the elongation at break increased and the tensile yield strength decreased, indicating excellent mechanical toughness^[8]. Zhang et al. investigated the thermal ageing properties of IPP/POE blends. The results showed that with pure IPP as a reference, the elongation at break of IPP/POE increased by 54.8% and the modulus of elasticity decreased by 42.5%. The introduction of POE had a positive effect on the

improvement of the mechanical properties of IPP^[9]. Hu et al. selected IPP, ethylene-propylene copolymer (PEC) and impact PP copolymer (IPC) as three matrices to graft styrene (St), respectively, and analyzed the thermal, mechanical and dielectric properties. The experimental results showed that the dielectric properties of these grafted modified PPs all showed improvement, but the degree of improvement was different^[10]. Huang et al. used aqueous grafting technology to graft styrene onto the PP molecular chain, tested the comprehensive electrical, thermal, mechanical and other properties, and compared the grafted material with XLPE^[11]. The results showed that grafted PP with excellent thermal properties, sufficient mechanical flexibility, and significant electrical properties, grafted polypropylene is very suitable for use in high-voltage cable systems. From the above studies, it can be concluded that the elastomer delivers low elastic modulus and excellent toughness, which is good for strengthening the mechanical performance of PP, while the low breakdown strength of elastomer degrades the insulation of the composites. Grafting modification can improve the insulation of PP, but almost no positive effect on the mechanical toughness. Thus, the combined modifications of blending and grafting may be conducive to balance the electrical and mechanical properties of polypropylene.

In this work, we propose to balance the mechanical and electrical properties of PP using combined modifications. The optimized blending content of IPP and BPP is determined and the IPP/BPP blends serve as the base materials. Furthermore, POE and styrene-grafted POE (POE-g-St) have been incorporated into IPP/BPP blends respectively to enhance both the toughness and insulation. The overall performances

of IPP/BPP/POE, IPP/BPP/POE-g-St composites have been systematically investigated. The results show that the electrical, mechanical and thermal properties have been significantly improved in the IPP/BPP/POE-g-St composites due to the synergistic effects of blending, elastomer POE and grafting styrene. The modified PP exhibits great potential to be used in the high-voltage cable insulation.

2 Experimental Procedure

2.1 Raw Materials

IPP and four types of BPP named as BPP-1, BPP-2, BPP-3 and BPP-4 were purchased from Sinopec. IPP is isotactic polypropylene formed by homopolymerisation of propylene monomer; the four types of BPP are all formed by block copolymerization of ethylene and propylene monomers and contain a rubber phase in the molecular chain. Elastomer POE (LC700) is produced by LG Corporation of South Korea.

Styrene (St) was used as the grafting monomer of the elastomer POE, and dicumyl peroxide (DCP) was used as the grafting initiator. Other reagents required for the experiment included polystyrene, xylene, and acetone, all of which were purchased from Innochem. Antioxidant 1010 produced by Macklin Company was selected and added at 0.3wt% of PP.

2.2 Preparation process

All materials were melted at 180°C in an internal mixer, made into pellets after cooling, hot-pressed at a pressure of 15 MPa for 20 minutes in a flatbed vulcanizer, and then cooled for 10 minutes until it reaches room temperature, and pressed into

samples of different thicknesses according to test requirements. IPP/BPP and IPP/BPP/POE are prepared by mixing the raw materials in proportion and then putting them into an internal mixer for melt blending. The IPP/BPP/POE-g-St is prepared by first preparing POE-g-St, and then melt blending the grafted product with IPP and BPP. The POE-g-St material preparation process is as follows: POE, St and DCP are physically mixed in proportion and then put into a mixer to be melt-grafted at a certain temperature and rotor speed. The grafted product is taken out after the reaction stops at a predetermined time. In order to eliminate the influence of impurities such as unreacted monomers and homopolymers, a portion of the grafted product was heated and refluxed in xylene at 100°C for 40 minutes to completely dissolve it. Pour it into the acetone solution while it is still hot, stir it continuously to get flocs, and filter the flocs with a Buchner funnel. The filtered material was added back into xylene, and this process was repeated three times. The treated product was then placed in an oven at 70°C and dried for 24 hours to a constant weight, thereby obtaining a purified grafted product. In this paper, three grafted products with different grafting contents were prepared. The ratios of POE, St and DCP were 40: 1: 0.2, 40: 2: 0.4 and 40: 4: 0.6 (phr), respectively. The grafting rate was calculated based on infrared scanning, which will be introduced in detail below.

The naming scheme of all the materials in this paper is as follows: IPP/BPP materials were mixed in different ratios as shown in Table S1, numbered from #1 to #16; The "x" in IPP/BPP/POEx composite material represents the weight percentage of POE. For example, IPP/BPP/POE20 means that POE accounts for 20wt%. The "y"

in IPP/BPP/POE-g-Sty represents the grafting content of styrene on the elastomer POE. For example, IPP/BPP/POE-g-St0.83wt% means that POE with a grafting content of 0.83wt% styrene in IPP/BPP blends.

2.3 Structure and performance characterization

Scanning electron microscope (SEM) was used to characterize the morphology structure of the prepared materials. The specimens were pressed into a thickness of 1 mm, immersed in liquid nitrogen for 30 s, and then quenched. IPP/BPP/POE specimens need to be etched in n-heptane solution after quenching to remove the elastomer POE to be used to observe the dispersion state of the elastomer. The detailed steps are as follows: The quenched sample was immersed in n-heptane solution and ultrasonicated for 30 min at 50°C. After the ultrasonication, the sample was placed in clean water and cleaned by ultrasonication for 30 min. Finally, the sample was taken out and dried. All specimens need to be sprayed with gold on the cross section before testing, so that the specimen is electrically conductive and the morphology of the specimen can be observed under SEM.

X-ray diffraction(XRD) experiments are used to characterize the crystal phase of composite materials. A Cu target was used as the X-ray emission source with a wavelength of 0.153 nm, an operating voltage and current of 40 kV and 40 mA, respectively, a step size of 0.05°, and a scanning range of 0° to 30°.

Differential scanning calorimetry (DSC) was used to test the melting and crystallization behavior of the materials. The test process is carried out in a nitrogen (N₂) atmosphere, and the sample quality is controlled at 7±0.5mg. It is packaged in an

aluminum crucible. The test procedure is as follows: The temperature was increased from 25°C to 200°C at a rate of 10°C/min and held at a constant temperature for 2 minutes to eliminate thermal history. The temperature was then lowered from 200°C to 25°C at a rate of 10°C/min, and the data of the process was collected to draw a crystallization curve; thereafter, the temperature was raised from 25°C to 200°C at a rate of 10°C/min, and the data of the process was collected to draw a melting curve.

The calculation formula of crystallinity is as follows:

$$X_C = \frac{\Delta H}{\Delta H_0} \times 100\% \quad (1)$$

where X_C is the crystallinity, ΔH is the melting enthalpy of sample, ΔH_0 is the melting enthalpy for PP with 100% crystallinity and is selected as 209 J/g. In this paper, fourier transform infrared spectroscopy (FTIR) was used to characterize the chemical structure of the grafted composites. All samples were dried in an oven at 60°C for 12 h before testing, with a test beam range of 4000-400 cm⁻¹ and a resolution of 2 cm⁻¹.

An electronic universal testing machine was used to test the mechanical properties of the samples, and the data on elongation at break and tensile strength were used to characterize the mechanical properties of the materials. The stretching was performed at a rate of 50 mm/min at room temperature. The temperature range of the dynamic thermomechanical (DMA) test was -80°C~160°C, the linear heating rate was 3°C/min, the double cantilever clamp tensile mode was adopted, and the vibration frequency was fixed at 1 Hz.

Asymmetric circular electrodes were used to conduct DC breakdown tests on the

samples. The electrodes and samples were immersed in dimethyl silicone oil. The voltage of the sample was linearly increased at a rate of 1kV/s until the sample broke down. The breakdown voltage at this time was recorded. The calculation formula of breakdown field strength is:

$$E_0 = \frac{U_0}{d} \quad (2)$$

where E_0 is the breakdown field strength of the sample, U_0 is the voltage when the sample breaks down, d is the thickness of the sample.

The data were statistically analyzed using Weibull distribution and plotted into graphs.

The volume resistivity of the sample was tested using a three-electrode system. The platform was built independently by the laboratory and mainly includes a high-voltage DC power supply, a current-limiting resistor, a three-electrode system, an electrometer, a constant temperature box, etc. Before the test, aluminum electrodes were coated on both sides of the specimen surface and the volume resistivity of the specimen was tested at different temperatures under a DC field of 30 kV/mm. The volume resistivity is calculated as follows:

$$\rho_v = \frac{U}{I_c} \cdot \frac{\pi(d+g)^2}{4L} \quad (3)$$

where U is the test voltage, I_c is the conductance current value, d is the diameter of the measuring electrode, g is the distance between the measuring electrode and the guard electrode, L is the thickness of the sample.

The space charge test is completed by the pulse electro-acoustic (PEA). The test

temperature is 30°C, and the reference waveform is obtained by polarization for 5 minutes under an electric field of 3kV/mm. Polarize for 30 minutes under 40kV/mm electric field, then short-circuit for 30 minutes, and test the change of space charge density over time. According to the space charge distribution in the depolarization process, the average space charge density in the depolarization process is calculated.

The calculation formula is as follows:

$$Q(t) = \frac{1}{D_1 - D_0} \int_{D_0}^{D_1} |Q_0(x, t)| dx \quad (4)$$

where D_0 and D_1 are the cathode and anode positions during the sample space charge test, Q_0 is the depolarization space charge density inside the sample, t is the depolarization time.

3 Results and discussion

3.1 Optimal blending ratio of IPP/BPP blends

It is well-known that isotactic polypropylene (IPP) can't be used as cable insulation directly due to its high elastic modulus and high crystallinity. While block polypropylene (BPP) synthesized by copolymerization of ethylene and propylene that contains a rubber phase and has a lower Young's modulus. Although the breakdown strength of BPP is generally lower than that of IPP, which is still much higher than that of thermoplastic elastomer^[12]. Thus, blending IPP with BPP at an appropriate content can reconcile the toughness and insulation of the blends. To determine the optimal blending ratio of IPP and BPP, four kinds of BPP are blended with IPP at different blending ratios, the optimal blending ratio is achieved in the consideration of the overall performance.

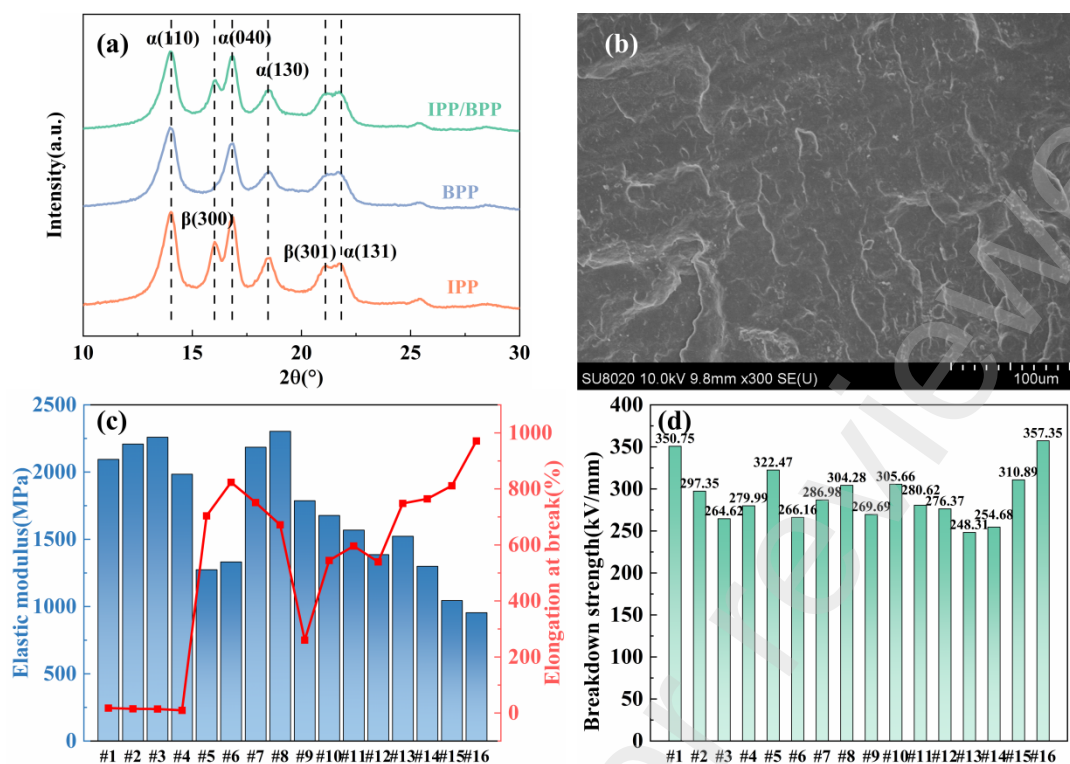


Fig. 2 Microstructure and macroscopic properties of IPP/BPP blends. (a) XRD patterns. (b) SEM image. (c) Elastic modulus and elongation at break. (d) Breakdown strength.

Fig.2 displays the microstructure, mechanical and electrical properties of IPP/BPP blends. XRD patterns show that the crystalline IPP are composed of α and β phases, while only α phase can be observed for pure BPP, as given in Fig.2(a). There are five characteristic peaks for the IPP/BPP blends, at $2\theta=14.0^\circ$, 16.8° , 18.6° , 21.0° , 21.8° corresponding to $\alpha(110)$, $\alpha(040)$, $\alpha(130)$, $\beta(301)$, $\alpha(131)$ crystal planes, respectively, which indicates that no chemical reaction or other impurities phases are incorporated in the IPP/BPP blends. It can be seen from SEM image as provided in Fig.2(b) that IPP and BPP are highly compatible and no obvious interface separation can be observed, presenting uniform and compact microstructure.

The overall properties including stress-strain curves, Weibull distribution and DSC curves for the 16 specimens blending IPP with four kinds of BPP and different blending ratio are shown in Fig. S1-S3, the key parameters are shown in Table S2-S4

and plotted in Fig. 2(c). It can be seen that all #1-#4 specimen exhibit ultrahigh elastic modulus around 2000 MPa and low elongation at break, meaning that BPP-1 has no positive effect on improving the mechanical toughness of IPP. This phenomenon maybe caused by the high molecular weight of the blends, making it difficult to be extruded and deformed. For the specimen #5-#8, the elongation at break changes a little but elastic modulus increases with the increase of blending ratio of BPP-2, which also can't improve the mechanical property as expected. In contrast, the specimen #9-#16 almost shows an increase in elongation at break and a decrease in elastic modulus, demonstrating that the mechanical toughness is significantly improved by blending BPP-3 and BPP-4. For instance, #16 sample possesses the lowest elastic modulus of 954.51 MPa and the largest elongation at break of 970.98%, which can be attributed to the ethylene chain segments in the molecular chains of the block polypropylene that promote the relative slippage of the molecular chains under external forces, improve the flexibility of the molecular chains^[13].

To further screen out the optimal BPP and its blending ratio, the electric breakdown strength of all blends is measured and shown in Fig. 2(d) and Fig. S3. It can be found that the breakdown strength of the blends with BPP-1 and BPP-2 slightly decreases with blending content increasing. This phenomenon maybe caused by the ethylene chain segments in BPP increases the loose regions and disrupts the tight structure of the molecular chains, providing the migration pathway for free-electrons and reducing the breakdown strength. Comparing to the highest breakdown strength of 305.66 kV/mm for #10 IPP/BPP with BPP-3, it is worth noting that the

breakdown field strength for #13 to #16 blends with BPP-4 shows a gradually increasing trend with the increase of BPP blending content, achieving the maximal value of 357.35 kV/mm. It has been shown that organic molecules with the same or similar groups have stronger van der Waals forces when they mixed together^[14,15]. It is possible that the higher propylene content in block polypropylene means that the molecular chains are more compatible and more tightly bound after blending IPP with BPP-4, resulting in a higher breakdown field strength^[16].

Additionally, DSC results are shown in Fig. S3 in the supporting information. Because polypropylene cables insulation always works at long-term elevated temperature higher than 70°C, which requires the insulation with a high melting temperature and excellent thermal stability. It can be seen that the melting temperature for all the specimens are above 160°C, except #16 that possesses a relatively low melting temperature of 158.38°C. The crystallinity can be also calculated from the DSC curves, it shows that the crystallinity degrees of #1 to #12 specimen are from 41% to 46.78%. The much higher crystallization corresponds to the poorer mechanical toughness and excessive stiffness. In contrast, the crystallinity degrees of #13 to #16 decreases with the increase of BPP blending content, and the decrease in crystallinity represents a more disordered arrangement of molecular chains, which is softer and has better mechanical properties. Combining the above parameters of mechanical, electrical and thermal properties, #15 with 40 wt.% BPP-4 is finally selected as the base blends for the next part of the study.

3.2 Microstructure and properties of IPP/BPP/POE

As aforementioned, blending BPP at an appropriate content can reduce the elastic modulus of IPP and increase the elongation at break, however, the mechanical toughness of IPP/BPP blends is still inferior to XLPE, restricting the real application of PP cable insulation. Therefore, we further propose to improve the mechanical property of IPP/BPP blends by incorporating POE elastomer, the IPP/BPP/POE composites with 20wt.%~50wt.% POE are prepared and their microstructure, electrical, mechanical and thermal properties have been fully investigated.

The cross-sectional SEM images of IPP/BPP/POE composites are shown in Fig. 3(a1)-(a4). It shows that IPP/BPP blends are served as mother-phase, while amount of voids left after etching the elastomer POE, indicating the so-called “sea-island” structure of IPP/BPP/POE. As the elastomer POE content increases, the diameter size of the voids left increases accordingly from a few micrometers to about ten micrometers. The size and distribution of POE, especially the interface structure, have a deep impact on the mechanical and electrical properties of IPP/BPP/POE. To study the effect of POE on the crystal phases of IPP/BPP, XRD patterns are employed and displayed in Fig. 3(b). Comparing to that of IPP/BPP, it can be seen that the diffraction peak intensity corresponding to β crystal phases at 2θ of 16.0° decreases gradually with the increase of POE content^[17]. Meanwhile, the diffraction peaks intensity of $\alpha(130)$ and $\alpha(040)$ change dramatically. This means that the incorporation of POE elastomer has significant effect on the molecular chains conformation and the crystal structure of IPP/BPP, restricting the formation of β crystal phase and increasing α crystal phase along the (040) crystal plane.

The DSC curves are characterized for studying the melting and re-crystallization behaviors of IPP/BPP/POE and are shown in Fig. S4 and Fig. 3(c). It can be seen that a single endothermic peak and an exothermic peak are observed after incorporating POE elastomer. The re-crystallization exothermic peak moves towards the lower temperature and the width of the re-crystallization peak increases^[15], demonstrating that the introduction of POE elastomer exerts a detrimental influence on the crystallization process, impeding the crystallization. The reduction in crystallinity results in a more loosely arranged molecular chain structure. When impacted by external forces, the molecular chains are more susceptible to slippage relative to one another, which is beneficial to improving the toughness. Following the addition of the elastomer, the melting temperature of the blend remains largely unaffected, with the overall temperature remaining above 160°C.

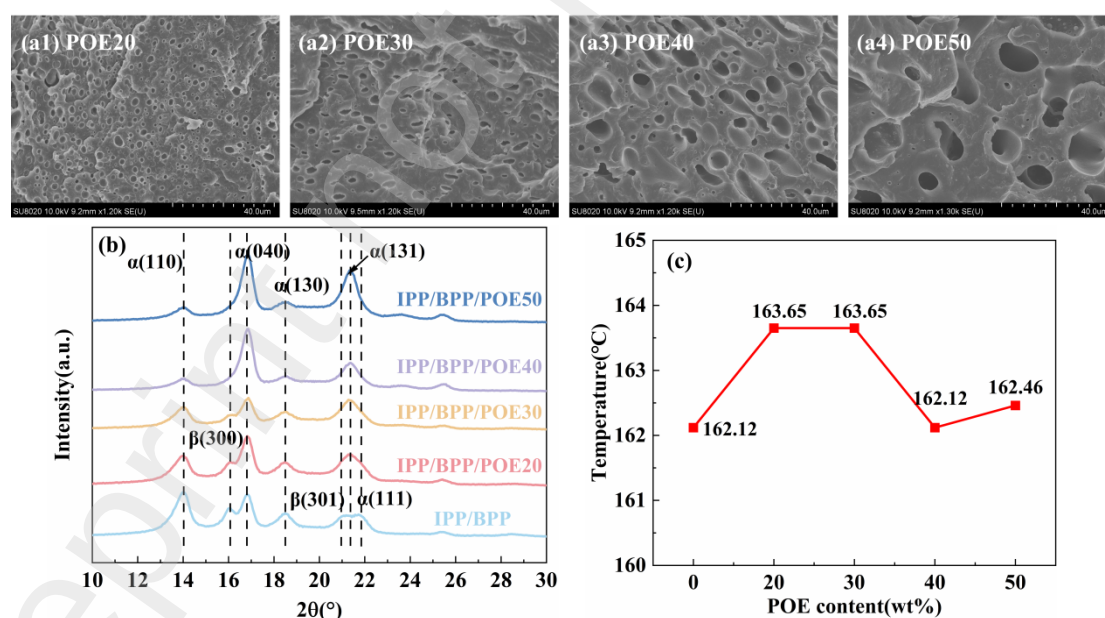


Fig. 3 Microstructure of IPP/BPP/POE composites at different POE contents. (a1)-(a4) SEM images. (b) XRD patterns. (c) Melting temperatures.

The stress-strain curves of IPP/BPP/POE are shown in Fig. S5, the key parameters are extracted and shown in Fig. 4(a). As the proportion of POE elastomer

increases, the elastic modulus decreases significantly, while the elongation at break increases, indicating that the incorporation of POE elastomer enhances the mechanical toughness. The long octene chain segments in POE exhibit a relatively low degree of rigidity, which forms a network that can act as a buffer, undergo a certain degree of deformation and absorb a proportion of the stretching energy when it is subjected to external tensile forces^[18]. Furthermore, the reduction in crystallinity can also facilitate the movement of molecular chains during stretching, thereby enhancing the elongation at break. The slight decrease in elongation at break for the composite with 50wt% POE, which may be attributed to the excess elastomer that increases the amorphous area and weakens the entanglement between the molecular chains^[19]. The tensile strengths of IPP/BPP/POE are lowered with POE increasing due to the low elastic modulus and tensile strength of POE.

DMA testing was also performed to investigate the thermal-mechanical property of IPP/BPP/POE, the results are shown in Fig. 4(b)-(c). It can be seen that the storage modulus decreases with increasing temperature, which is consistent with the relaxation characteristics of polymers. The rapid reduction happens around 0°C that corresponds to the glass-transition temperature of pure PP. Additionally, the energy storage modulus of IPP/BPP/POE composites is markedly inferior to that of IPP/BPP, the lowest energy storage modulus reduces to approximately 580 MPa at 25°C, which aligns with the reduction of elastic modulus in stress-strain curve. There are two loss peaks observed at -30°C and 0°C, which correlate with the glass transition temperature of the elastomer and polypropylene, respectively. When the temperature

falls enough low, the molecular chain segments are in a frozen state and difficult to move. In contrast, the molecule chains become moveable when the temperature increases^[20]. The energy of molecular thermal motion rises, resulting in the excitation of molecular chain segments and transforming into highly elastic state. During the transition processes, the molecular chain experiences a degree of slippage, accompanied by an increase in the friction that must be overcome. This leads to an elevation in the level of internal friction, reaching a maximum value at the glass transition temperature.

Besides the mechanical property, the electrical insulation is also very important for POE modified IPP/BPP. Fig. S6 shows the Weibull distribution of characteristic breakdown field strength of IPP/BPP/POE measured at different temperatures, and the extracted breakdown strength values are provided in Fig. 4(d). Comparing to IPP/BPP, the incorporation of POE decreases the electrical insulation due to the intrinsic lower breakdown strength of POE than IPP/BPP. For IPP/BPP/POE, the breakdown strength shows a trend of first decreasing and then increasing with the increase of POE elastomer content. It has been demonstrated that as the POE content reaches 40wt%, there is an increase in the breakdown field strength. This phenomenon is different from previous studies that the breakdown strength decreases with the increase of elastomer content, which can be attributed to the fact that the binary base matrix in this study contains ethylene segment in BPP component. Because the ethylene segments are presented in both BPP and POE elastomer, the similar molecular groups are beneficial to enhance the van der Waals force between

base matrix and POE, resulting in a more compact and tightly entangled molecular structure, which in turn increases the breakdown field strength. Additionally, the breakdown strength demonstrates a decline at 60°C and 90°C comparing to that 30°C. As the increase of measured temperature, the thermal-excited electrons obtain more energy and migrate, which results in an increased probability of electron bombardment of the molecular chain, leading to the degradation of molecular chain and the reduction of electric breakdown strength^[21,22].

DC volume resistivity at different temperatures under an electric field of 30 kV/mm is shown in Fig. 4(e). As the proportion of POE elastomer increases, DC volume resistivity firstly decreases, then increases and finally decreases again, exhibiting the similar variation trends as the breakdown strength of IPP/BPP/POE. Of course, DC volume resistivity for IPP/BPP/POE is lower than that of pristine IPP/BPP. As above-mentioned that the crystallization degree decreases and the molecular chains are more loosely arranged with the increase of POE content, which leads to an increase in the volume of the amorphous region and accelerates the carrier migration. When the elastomer content increases to 40wt%, the volume resistivity increases. This is because a higher trap density appears at the interface between POE and polypropylene, which can capture a large number of carriers, thereby reducing the conductive current and increasing the volume resistivity^[19,23]. When the POE content increases to 50wt%, the volume resistivity decreases slightly. The SEM results indicated that an increase in POE content led to the formation of interconnections between certain elastomer, resulting in a reduction in the interface density within the

material, which is believed to be the reason for the slight decrease in volume resistivity. As temperature increase in 30°C, the volume resistivity decreases by about one order of magnitude. It is believed that at high temperatures, the energy obtained by the carriers increases, and the traps generated at the interface are relatively shallow, rendering them unable to capture the carriers migrating at high temperatures. Especially, impurities may dissociate at high temperatures and participate in the process of ion conduction.

The space charge density distribution curves with time are shown in Fig. S7(a)-(e). No obvious charge accumulation at the cathode and anode can be observed, while a few charges accumulate in the bulk of IPP/BPP. It may be caused by that the impurities (antioxidants, etc.) in the polymer are ionized under the electric field, but the quantity of the charges is very low. On the contrary, the incorporation of POE elastomers result in hetero-charges around the cathode and anode for IPP/BPP/POE. These charges subsequently migrate towards the interior of the specimen over time^[24,25]. The introduction of the POE elastomer in IPP/BPP has formed two-phases interfaces, which also induce charge traps that can capture the carriers. A substantial amount of charge accumulates give rise to discernible charge packets that distort the local electric field even induce electric breakdown, and this phenomenon becomes more pronounced as the elastomer content increases^[19,24,26].

The results of the spatial charge density changing with time calculated based on the spatial charge distribution during the depolarization process are illustrated in Fig. 4(f). For the IPP/BPP, the charges decay rapidly within 50 s and then tend to be

stable. During the first 50 seconds, carriers only need to overcome a very low potential barrier to detrapping from shallow traps. After 50 seconds, the average charge density tends to be stable and no longer decays, because the charges in the deep traps have to overcome a very high potential barrier, which is manifested as a slow decay rate of the space charge density [19]. The charge dissipation process of IPP/BPP/POE is similar to that of IPP/BPP. While a large amount of charge still remains in the polymer after the charge dissipation under stable state. This phenomenon indicates that there are deep traps at the interface that make it difficult for the charges to escape, the charges accumulation would lead to the electric field concentration and distortion, degrading the insulation property. Although the elastic modulus decreases from 1044.74 MPa to 382.61 MPa, the breakdown strength drops from 310.89 kV/mm to 262.59 kV/mm for IPP/BPP and IPP/BPP/POE40, respectively, demonstrating that the improved mechanical toughness is achieved at the expense of electrical insulation property. In this case, the styrene-grafted POE (POE-g-St) is proposed to balance the mechanical and electrical properties for IPP/BPP/POE40.

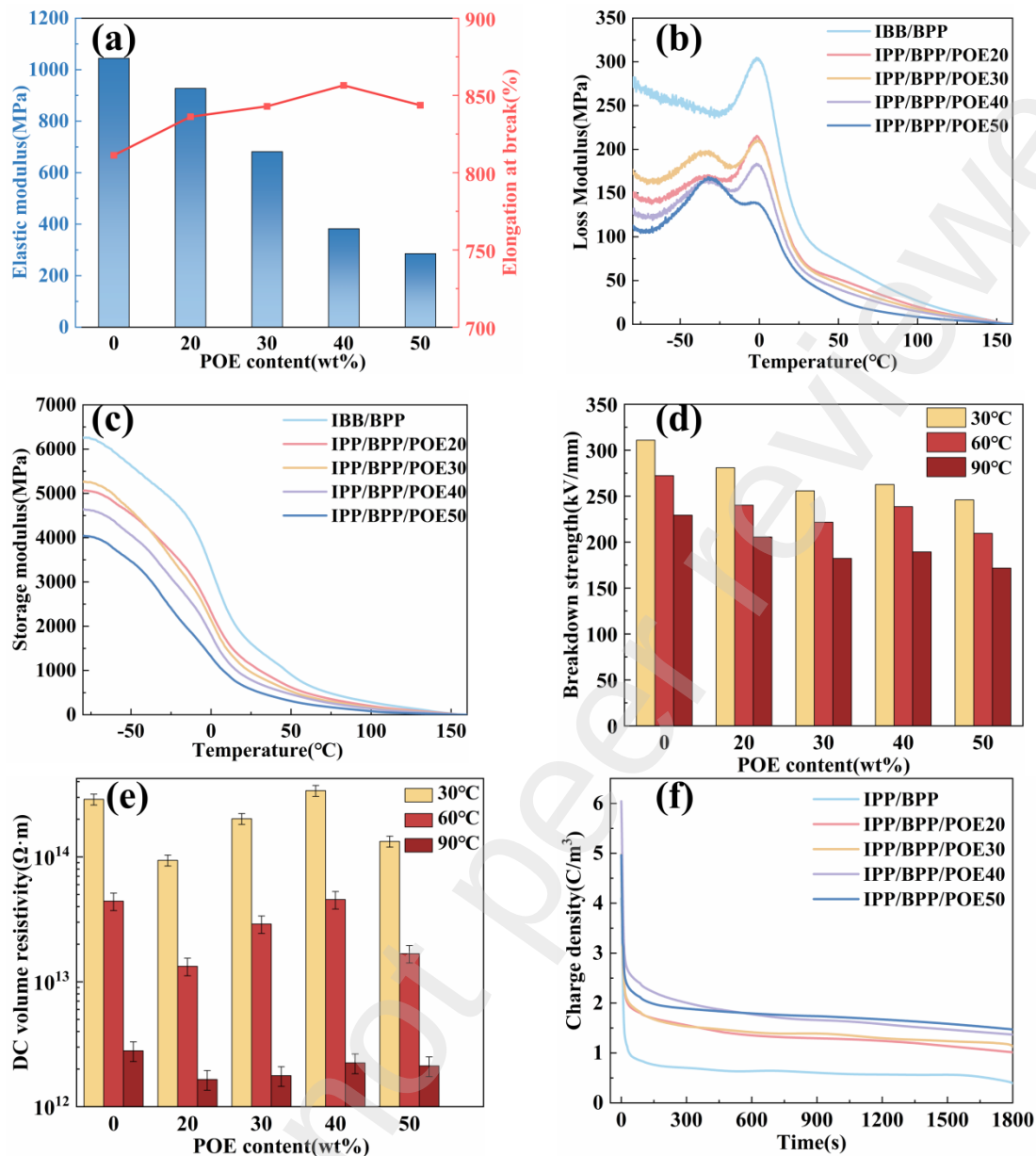


Fig. 4 Macroscopic properties of IPP/BPP/POE with different POE contents. (a) Elongation at break and elastic modulus. (b) Energy storage modulus. (c) Loss modulus. (d) Electric breakdown strength. (e) DC volume resistivity. (f) Average charge density

3.3 Microstructure and properties of IPP/BPP/POE-g-St

The schematic diagram of the grafting reaction process for styrene-grafted POE (POE-g-St) is shown in Fig. S8. To prove the successfully grafting St on POE molecular chains, the FTIR spectra are performed and the results are shown in Fig. 5(a). It can be seen POE-g-St shows an additional absorption peak at 700 cm⁻¹ corresponding to the out-of-plane vibration of the C-H bond of substituted benzene,

whereas no absorption peak at 700 cm^{-1} can be observed for pure POE, proving that St has been successfully grafted onto POE. In addition, the grafting rate of St onto POE can be quantitatively analyzed according to the Lambert-Beer law^[27]. However, since the characteristic absorption peaks of the grafted product and the homopolymer in the blends are completely consistent, it is hard to distinguish the absorption peaks as well as to calculate the grafting rate of POE-g-St. Herein, we propose an equivalent method to calculate the grafting rate by using the blend of polystyrene (PS) and POE. The POE and PS are blended in an internal mixer with different weight percentages and the FTIR are measured as shown in Fig. S9. The long-chain ethylene group - $(\text{CH}_2)_n$ - ($n \geq 5$) at 720 cm^{-1} and characteristic peak at 700 cm^{-1} are selected as the internal standard peak of POE and POE-g-St, respectively. The baseline method was used to calculate the characteristic absorption peak areas $A(720)$ and $A(700)$ of POE and POE-g-St. With $A(700)/A(720)$ as the ordinate and PS content as the abscissa, the working curve is obtained by linear fitting and univariate linear regression method, as shown in Fig. 5(b). According to the weight percentages of St used in this study, the grafting rates of St can be calculated as 0.83wt%, 1.21wt% and 1.76wt%, respectively.

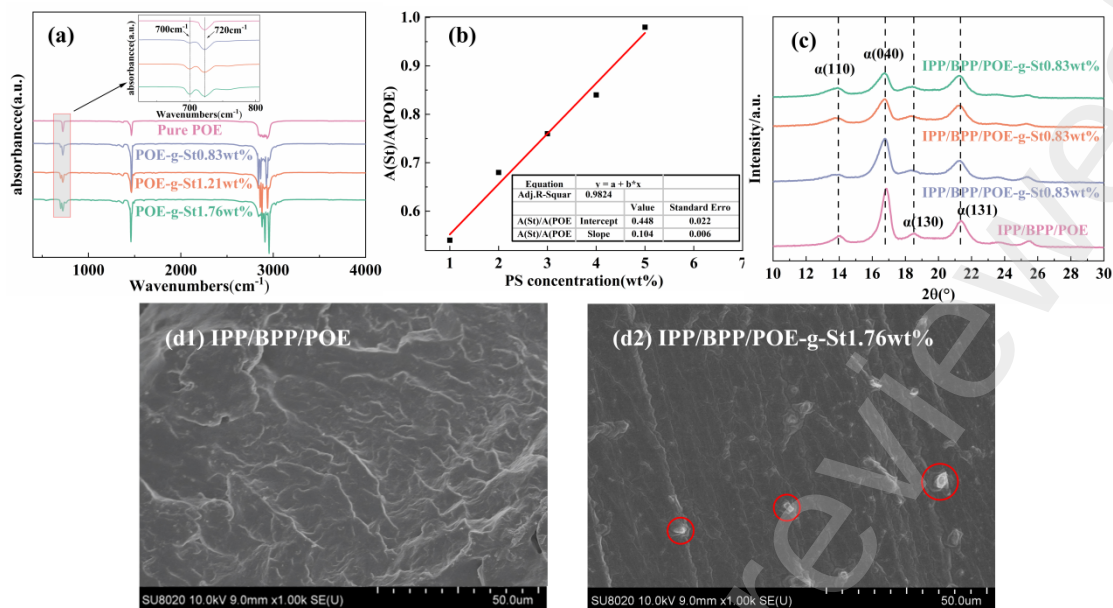


Fig. 5 Microstructure of POE-g-St and IPP/BPP/POE-g-St with different St contents.(a) FTIR spectra. (b) Linear fitting plot for calculating grafting rate. (c) XRD patterns. (d1)SEM image of IPP/BPP/POE (d2) SEM image of IPP/BPP/POE-g-St1.76wt.% .

XRD patterns of IPP/BPP/POE-g-St are shown in Fig. 5(c). It can be found that four diffraction peaks at $2\theta=14.0^\circ$, 16.8° , 18.6° , 21.2° correspond to the $\alpha(110)$, $\alpha(040)$, $\alpha(130)$ and $\alpha(111)$ crystal planes of IPP/BPP matrix. Although the crystal phases of IPP/BPP/POE are similar to IPP/BPP/POE-g-St, the diffraction peak intensity of $\alpha(040)$ decreases, demonstrating that POE-g-St restrict the molecular chains of IPP/BPP arrange along the $\alpha(040)$ crystal orientation. The cross-sectional SEM images of IPP/BPP/POE and IPP/BPP/POE-g-St1.76 wt.% are illustrated in Fig.5(d1)-(d2). Comparing to IPP/BPP/POE, the globular particles with red circle corresponding to St are visible that may be induced by the self-polymerization, curling and clustering. The agglomeration of St just occurs at high grafting content of St, which may negatively affect the electrical properties. It was reported that St and the interfaces between St and polymer matrix can capture carriers, thus enhancing the electrical insulation properties^[28]. DSC curves as given in Fig. 6(a) and Fig. S10

shows that the slight increase in the melting temperature of IPP/BPP/POE-g-St, which is attributed to the high heat-resistance of the introduced benzene ring structure after grafting St. The re-crystallization temperature first decreases and then increases, which may be induced by the heterogeneous nucleation effect due to grafting styrene [29]. In addition, the benzene ring on styrene has a higher rigidity and an obvious steric hindrance, which hinders the movement of the molecular chain segments [30], which may limit the growth of lamellae during the crystallization process, resulting in a decrease in the crystallinity, as provided in Table. S8.

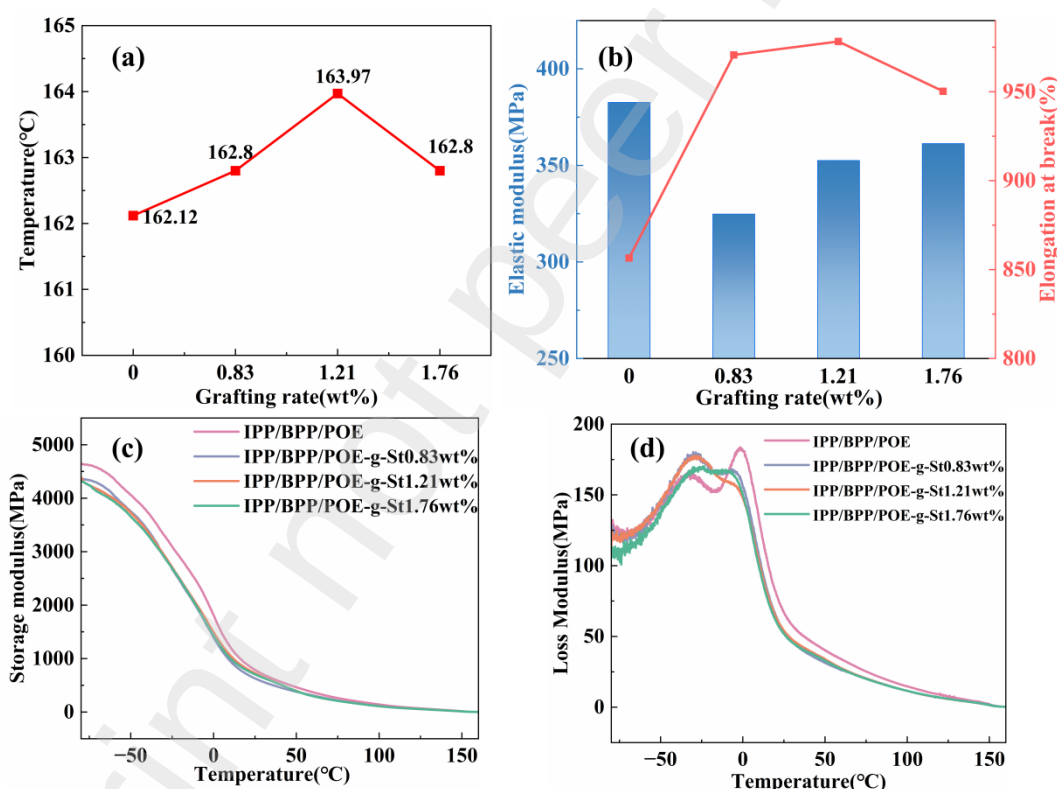


Fig. 6 Macroscopic properties of IPP/BPP/POE-g-St with different grafting rates. (a) Melting temperatures. (b) Elongation at break and elastic modulus. (c) Energy storage modulus. (d) Loss modulus.

The stress-strain curves and the key mechanical parameters of IPP/BPP/POE-g-St are shown in Fig. 6(b) and Fig. S9. The elastic modulus of IPP/BPP/POE is 382.61 MPa, which first decreases then increases with the grafting rate increasing. The elongation at break before grafting is 856.50% for IPP/BPP/POE, while

IPP/BPP/POE-g-St possesses the prominent increase in the elongation at break, the maximal elongation at break of 978.22%, an increase of 14.2%, has been obtained for IPP/BPP/POE-g-St1.21wt%. The mechanism of the enhanced toughness is that the POE elastomer acts as a stress concentration point to absorb energy and inhibit the formation of "silver lines", thereby increasing elongation at break. Additionally, the introduction of styrene units improves the dispersion of the POE elastomer in the matrix, beneficial to inhibiting the formation of defects and increasing elongation at break as well. Meanwhile, the molecular chain spacing becomes larger and more susceptible to slippage due to the site-resistance effect of the introduced styrene units, leading to a reduction in tensile strength^[31]. Of course, the decreased crystallinity means that the molecular chain arrangement more disordered and loose, resulting in the reduced tensile strength.

Fig. 6(c)-(d) illustrates the energy storage modulus and loss modulus of IPP/BPP/POE-g-St. Comparing to IPP/BPP/POE, the energy storage modulus slightly decreases with the increase of St grafting rate, which may be caused by the steric effect and the inter-molecular spacing becomes larger, resulting in a weakening of the inter-molecular forces. The glass transition temperatures of POE elastomer and polypropylene appear around -30°C and 0°C, respectively^[32]. It can be seen from Fig. 6(d) that the loss modulus is mainly concentrated in the elastomer, which is obvious difference from IPP/BPP/POE40 without grafting St. This phenomenon is mainly attributed to the introduction of styrene units with higher stiffness that makes the relaxation of the molecular chains become more difficult, the sliding of the molecular

chains is restricted and the collision and friction of the molecular chains are increased, resulting in the enhanced loss modulus.

The space charge density versus time distribution curves of IPP/BPP/POE-g-St are shown in Fig. S12(a)-(c). It can be seen that the space charge is deeply affected by the grafting rates of St, a few homo-charges appear at both the cathode and anode for IPP/BPP/POE-g-St 0.83 wt%, which origin from the electrode injection and move towards the bulk of the specimen with time. For IPP/BPP/POE-g-St 1.76 wt%, the hetero charges appear near the cathode and a large amount of homo charges accumulate near the anode, which root in the ionized impurities and electrode injection respectively. There is almost no space charge accumulation near the cathode, anode as well as the bulk for the sample with a graft content of 1.21 wt% St^[33]. The mechanism of suppressing space charge accumulation can be explained by the trap theory^[34]. Fig. 7(a) illustrates the average charge density with time for the short circuit process. Compared to the ungrafted IPP/BPP/POE, the charge decay rates of IPP/BPP/POE-g-St with different grafting contents are significantly slower at the beginning (~300s) and the decay rate stabilize after about 300s. Once the grafting content reaches 1.76 wt%, the excess trap densities could accumulate amount of space charges, leading to difficulties in dissipating during short circuiting.

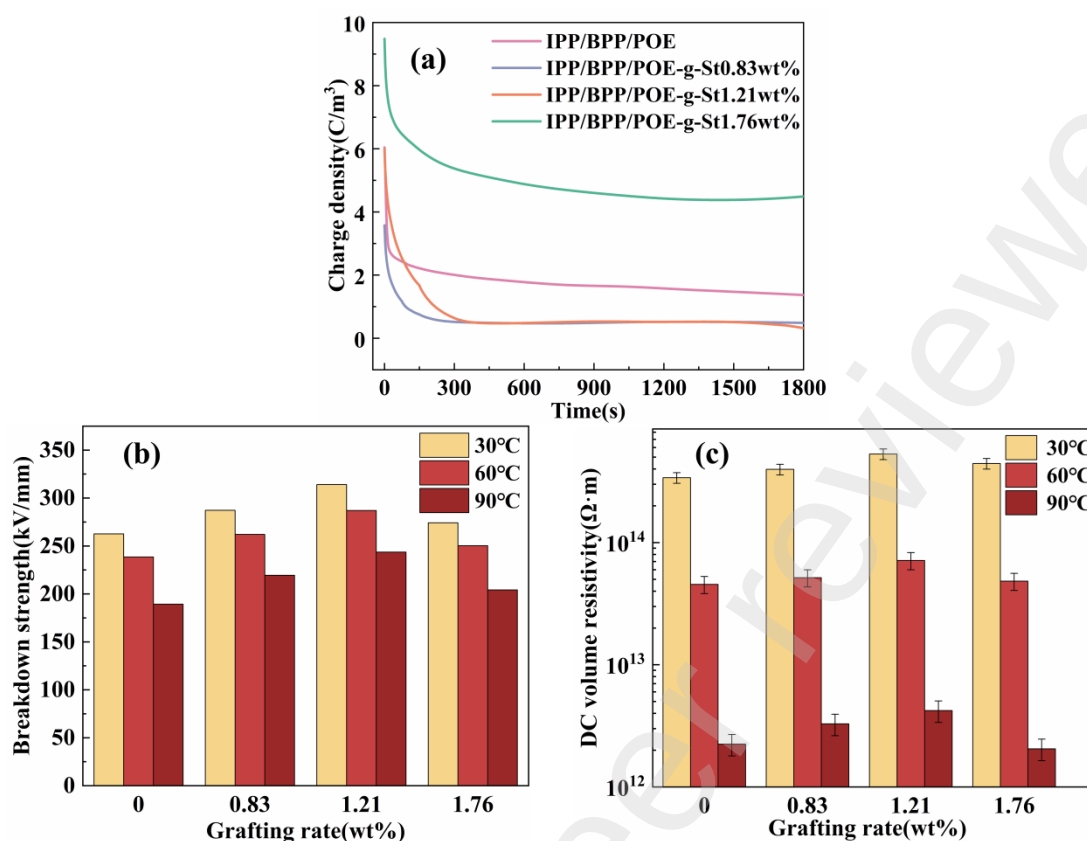


Fig. 7 Electrical insulation properties of IPP/BPP/POE-g-St with different St contents. (a) Average charge density. (b) Electric breakdown strength. (c) DC volume resistivity.

The breakdown strength and DC volume resistivity are measured for presenting the electrical insulation of IPP/BPP/POE-g-St. As shown in Fig. 7(b), the breakdown strength increases first and then decreases with the St grafting content increasing. The maximal breakdown strength reaches 314.02 kV/mm, 19.5% higher than IPP/BPP/POE, indicating that grafting St onto POE can improve the electrical insulation. St possesses a lower ionization potential and a higher electron affinity, which can be used as a voltage stabilizer to trap electrons or reduce electrons energy [35,36], preventing the high-energy electrons impact the molecular chain and thus increasing the breakdown field strength.

The volume resistivity of IPP/BPP/POE-g-St is shown in Fig. 7(c), which exhibits slightly higher than that of IPP/BPP/POE without grafting St. The volume

resistivity of IPP/BPP/POE-g-St1.21 wt% is elevated by 56.3% at 30 °C due to the deep traps effects induced by St. The grafted St acts as deep traps that can reduce the carrier mobility and increase the volume resistivity^[24,33,37]. Once the trap densities increase more significantly, the trapped carriers can migrate and hop between the adjacent traps, which can easily pass between adjacent traps by tunneling effect, leading to a decrease in DC volume resistivity^[30,38]. As the temperature increases, the thermal motion of the carrier increases and the ability of the trap to capture the carrier decreases, resulting in an increase in volume resistivity.

4 Conclusion

In summary, this study systematically investigates the effects of blending block polypropylene (BPP), incorporating polyolefin elastomer (POE) and styrene-grafted POE (POE-g-St) on the microstructure, mechanical, electrical and thermal properties of isotactic polypropylene (IPP). The results demonstrate that the overall performances, such as enhanced mechanical toughness and retentive electrical insulation are achieved for IPP blended with 40 wt.% BPP. POE elastomer is used to further reduce the elastic modulus and improve the toughness of IPP/BPP blends. Both of stress-strain curves and DMA results verify that as the proportion of POE elastomer increases, the elastic modulus decreases significantly, while the elongation at break increases, indicating that the incorporation of POE elastomer enhances the mechanical toughness of IPP/BPP blends. However, the breakdown strength and DC volume resistivity are degraded comparing to IPP/BPP blends without POE. In the consideration of the intrinsic breakdown strength of POE is much lower than

polypropylene matrix, so we further propose the usage of styrene-grafted POE (POE-g-St) as secondary phase, the results show that grafting modification could improve the electrical insulation of IPP/BPP/POE-g-St, maintaining excellent mechanical toughness.

Acknowledgement

This work was supported by the National Natural Science Foundation of China (No. 52277024).

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